

# ULTRON 2.7 Phase 11 Execution Report

Windows automation executor pack

|           |                                            |
|-----------|--------------------------------------------|
| Project   | ULTRON 2.7                                 |
| Phase     | Phase 11: Windows Automation Executor Pack |
| Workspace | Local workspace checkout: U2.7             |
| Status    | Implemented and verified                   |

## Executive Summary

Phase 11 adds a real Windows automation executor pack while keeping ULTRON 2.7 inside the existing typed-tool safety architecture. The assistant can now front practical laptop actions such as app launch, safe-root file work, notes, reminders, clipboard, screenshots, and optional volume/brightness adapters.

The LLM and planner still do not receive raw shell access. Every action remains user request -> brain/planner -> typed tool call -> schema validation -> policy gate -> executor adapter -> audit log -> response.

## Phase 11 Scope

- Add a Windows automation adapter module behind the executor.
- Restrict app launch to an approved alias registry.
- Block explicit file and folder paths outside configured safe roots.
- Require confirmation for medium-risk and high-risk actions.
- Keep destructive actions non-destructive and not implemented.
- Add a UI confirmation modal and confirmation-cancel API.
- Add a recent-audit endpoint for runtime inspection.
- Add tests and docs for the Windows pack.

## What Was Implemented

| Area            | File(s)                          | Result                                                                                                                                 |
|-----------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Windows adapter | src/ultron27/windows_executor.py | Typed helpers for app aliases, path opening, clipboard, screenshots, optional volume, optional brightness, and safe target validation. |

| Area      | File(s)                                              | Result                                                                                                                                                   |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Executor  | src/ultron27/executor.py                             | Routes Windows tasks through the adapter, blocks unsafe explicit paths, handles medium/destructive tools conservatively, and preserves dry-run behavior. |
| Policy    | src/ultron27/policy.py,<br>src/ultron27/models.py    | Adds permission levels and requires confirmation for medium-risk and high-risk commands.                                                                 |
| Planner   | src/ultron27/planner.py                              | Routes terminal commands to the medium-risk terminal tool instead of generic app launch.                                                                 |
| Audit API | src/ultron27/audit.py,<br>src/ultron27/web_server.py | Adds recent audit reading plus GET /api/audit/recent.                                                                                                    |
| Visual UI | web/index.html, web/styles.css, web/app.js           | Adds a confirmation modal for risky typed and voice actions, plus cancellation that clears pending voice confirmation.                                   |
| Tests     | tests/test_pipeline.py                               | Expands to 60 tests covering safe roots, alias resolution, confirmation blocking, destructive non-execution, and audit endpoints.                        |

## Permission Behavior

| Level                                  | Meaning                                                 | Phase 11 Handling                                                                           |
|----------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------|
| low_risk_allowed                       | Validated low-risk tool.                                | Allowed after schema validation and policy approval.                                        |
| medium_risk_confirmation               | Tool can affect local state or open powerful surfaces.  | Paused until explicit confirmation.                                                         |
| high_risk_confirmation                 | Tool has elevated risk.                                 | Paused until explicit confirmation.                                                         |
| destructive_blocked_or_not_implemented | Delete, move, rename, script, email, shutdown, restart. | Confirmation-gated, then executor returns not_implemented rather than destructive behavior. |
| blocked                                | Invalid, unsafe, or unknown execution request.          | Rejected before executor side effects.                                                      |

## Verification Results

| Check         | Command                                                                                                                                                                        | Result                                         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Unit tests    | \$env:PYTHONPATH='src'; python -m unittest discover -s tests                                                                                                                   | 60 tests passed.                               |
| Compile check | python -m py_compile<br>src/ultron27/windows_executor.py<br>src/ultron27/executor.py src/ultron27/policy.py<br>src/ultron27/web_server.py<br>scripts/run_phase11_windows_ui.py | Phase 11 Python modules compiled successfully. |

| Check              | Command                                                      | Result                                                                                                                                                          |
|--------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend syntax    | <code>bundled node --check web/app.js</code>                 | Frontend JavaScript parsed successfully.                                                                                                                        |
| Dataset evaluation | <code>python scripts/evaluate_dataset.py --split test</code> | Intent 0.9965, tool 0.9965, argument exact match 0.9579, confirmation 1.0. The intentional miss is Terminal now routing to the safer medium-risk terminal tool. |
| Safety regression  | <code>python scripts/evaluate_safety_regression.py</code>    | 362 cases, tool accuracy 1.0, policy action accuracy 1.0.                                                                                                       |
| API smoke          | Local HTTP checks                                            | Status, command confirmation, and audit recent routes returned expected payloads.                                                                               |

## Recommended Next Step

- Install optional Windows packages for real volume and brightness control if those features are needed on the target laptop.
- Add a small UI audit viewer panel on top of `/api/audit/recent`.
- Build a signed or user-reviewed app alias editor instead of editing JSON by hand.